

Tech Stack

- **Python:** All code, including AWS integration, logic, and web app, is written in Python.
- **Flask:** Handles HTTP requests, page routing, and serves the web interface where users input AWS credentials.
- **boto3:** Authenticates with AWS using user's credentials, calls IAM APIs, fetches users, groups, and policies, and returns structured data.
- **HTML/CSS:** Builds user-facing forms and tables for readability and user input on the web page.
- **Jinja2:** Allows Python variables and data to be inserted into HTML templates, displaying results to users.
- **Tabulate:** Formats permission and user data into organized tables for easier viewing within the browser or for conversion/export.
- **AWS IAM:** Stores and manages identity data that your application reads—it is not hosted by you but leveraged via APIs.
- **CSV:** Optional feature—exports permissions data to a CSV file for audits or offline review.
- **Pip:** Used in the terminal (`pip install flask boto3 tabulate`) to set up necessary Python packages.

Step	Operation	Key Function
1	Initialize IAM Client	<code>boto3.client('iam')</code>
2	List IAM Users	<code>iam.list_users()</code>
3	Get Managed Policies	<code>iam.list_attached_user_policies()</code>

4	Get Inline Policies	<code>iam.list_user_policies()</code>
5	Get Group Policies	<code>iam.list_groups_for_user() +</code> <code>iam.list_attached_group_policies()</code>
6	Handle Errors & Format Output	<code>try-except, tabulate, csv</code>
7	Integrate with CLI/Web	Flask or CLI display